

Supplementary Material: Regime-Conditioned Dynamical-EVT Diagnostics for Predictive Maintenance

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1 Extended Methods

The implemented state-space construction requires causal availability. If a delay embedding is used, every lagged coordinate is shifted so that Z_t contains only information observed at or before t . Scaling parameters, dangerous-region prototypes, and thresholds are fitted on the declared training or calibration partitions and reused without refitting on the test partition.

Runs declustering uses a non-negative integer run length m . If exceedance times are $e_1 < \dots < e_k$, then e_i and e_{i+1} belong to the same cluster when $e_{i+1} - e_i \leq m + 1$. Cluster severity may be summarized by size, duration, maximum observable, or integrated excess above threshold. These empirical quantities should not be equated with fault duration unless the sampling frequency, gap rule, and label definition justify that interpretation.

The inter-exceedance estimator is included because cluster inference can be sensitive to manual declustering choices [2]. The dynamical interpretation follows the hitting-time connection between returns to shrinking target sets and extreme-value laws [3]. The periodicity result motivates the caution that recurrence can reduce the extremal index without implying a specific mechanical cause [4].

Horizon-risk calibration is computed after freezing the event target. The current generated reliability diagram is a diagnostic based on ranked risk surrogates; a full probabilistic calibration claim would require held-out calibration data, uncertainty bands, and explicit acceptance criteria.

2 Additional Simulations

The simulation registry includes independent, periodic, noisy, mixture, missingness, and degradation families. Independent simulations check whether estimators approach the non-clustered target. Periodic simulations check whether the workflow detects clustered returns. Noisy simulations test whether finite perturbations mask recurrence, following the warning that observed noise can move finite samples toward apparent independence [1]. Regime mixtures and preprocessing perturbations test whether clustering can be generated by analysis choices rather than physical degradation.

The generated simulation asset used in the main paper is the reproducibility-scale artifact. It is sufficient for build validation, but it is not a replacement for the full registered experiment matrix. The supplement therefore treats simulation results as evidence about the implemented pipeline and estimator sensitivity, not as an exhaustive finite-sample theory.

2.1 Reference availability

Table S1 records, for every simulated family, how many configurations carry a reference extremal index, whether that reference is theoretical or numerical, and which clustering regime it falls in. Three families — noisy periodic dynamics, persistent shift, and regime mixture — have no established reference at any configuration: the two long-run estimates used to fix a numerical value disagree beyond the accepted tolerance, with discrepancies from 0.15 to 0.60 and no plateau in the block-based family. These are mixture and non-stationary processes for which a single extremal index is arguably not well defined, and they are reported as non-identifiable rather than assigned a value that one declustering convention happens to produce.

The complete estimator grid, carrying all thirteen reported quantities for every configuration and estimator, is distributed as `estimator_full_grid.csv` in the reproducibility package. Every value in the main-paper table is recomputable from it.

2.2 Sensitivity to perturbations

Figure S1 reports mean estimates under noise, missingness, and regime mixture. Added noise moves every estimator towards independence, which is the expected direction: perturbation breaks the deterministic recurrence that produces clustering, and the numerically established references for the noisy families sit between 0.94 and 0.98 accordingly. This is also why the high-replication grid, which runs at a single non-zero noise level, is reported in the main paper as evidence about near-independent processes.

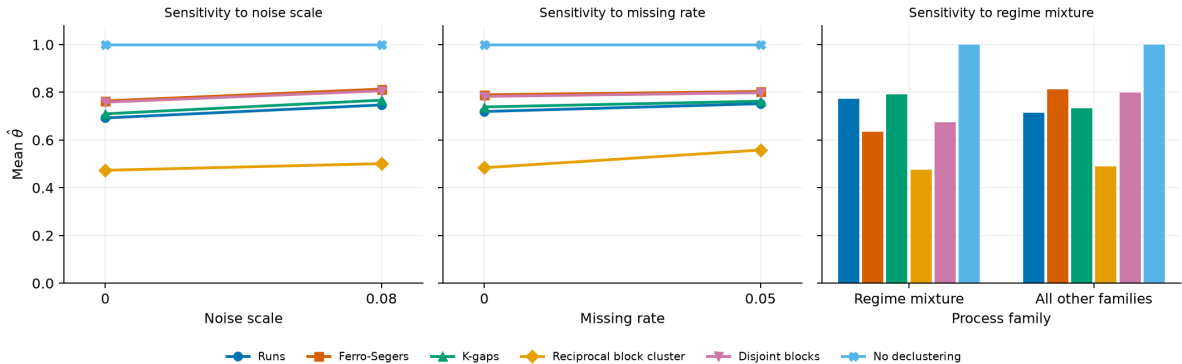


Figure S1: Estimator sensitivity to noise scale, missing rate, and regime mixture. The regime-mixture panel contrasts families rather than sweeping a parameter, since mixture is a property of the process and not a tuning knob.

Table S1: Reference availability and clustering regime for every simulated process family. Configurations without an established reference contribute no bias, RMSE or coverage evidence.

process	configurations	with reference	reference kind	reference theta	regime
burst	1	1	theoretical	1.000	near_independent
missingness					
cyclic	37	24	numerical	0.049, 0.062,	strongly_clustered
degradation				0.086, 0.175	
downsampling	1	1	numerical	0.981	near_independent
iid bounded	37	37	theoretical	1.000	near_independent
tail					
iid light tail	38	38	theoretical	1.000	near_independent
iid pareto	37	37	theoretical	1.000	near_independent
lagged	37	12	numerical	1.000	near_independent
multivariate					
logistic	37	31	numerical,	0.983, 0.984,	near_independent
nonperiodic			theoretical	0.992, 1.000	
target					
logistic	38	31	numerical,	0.500, 0.939,	moderately_clustered,
periodic			theoretical	0.940, 0.970	near_independent
target					
missing at	1	1	theoretical	1.000	near_independent
random					
noisy periodic	1	0	none		not
dynamics			established		established
persistent	1	0	none		not
shift			established		established
regime	37	0	none		not
mixture			established		established
smoothing	1	1	numerical	0.272	strongly_clustered

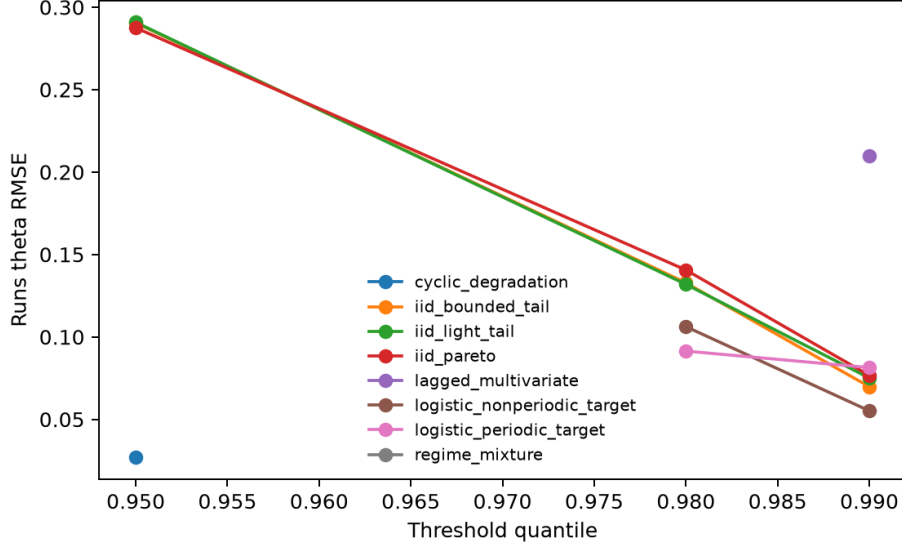


Figure S2: Supplementary copy of the generated simulation RMSE artifact.

3 Complete Event-Level Benchmark

Table S2 reports the complete held-out event-level benchmark under the shared alarm policy: every declared method on every compressor dataset, with detection status, event recall and precision, both alarm-burden axes, duplicate alarms and event utility. Table S3 records what each method actually computes, and Table S4 is the fairness audit.

Two properties of this benchmark are worth stating explicitly, because both were previously invisible.

First, method identity is enforced rather than assumed. Each method is declared by its score, its threshold rule and its event policy, and no two methods share all three. An earlier version of this benchmark reported nineteen methods of which nine produced bit-identical results, because a fallback assigned one shared score to any method lacking an explicit implementation; the classical peaks-over-threshold entry fitted no tail model and the declustering entries declustered nothing differently. The audit now reports any methods whose results coincide, so numerical agreement is visible as agreement rather than presented as independent evidence.

Second, alarm merging belongs to the detector. Four methods exist to vary declustering, so requiring one event conversion across the benchmark would make them indistinguishable by construction. The shared elements are the warning horizon, the matching tolerance, the matching rule, the operating exposure and the independent unit; the merge gap each method used is reported per row.

The complete benchmark including runtime, peak memory and configuration hashes is distributed as `event_benchmark_full.csv` in the reproducibility package; every value in the compact main-paper table is recomputable from it.

Table S2: Complete held-out event-level benchmark under the shared alarm policy: every declared method on every compressor dataset. Alarm burden is given as both interruption frequency (false alarms per operating day) and warning exposure (percentage of the test period under warning), because the two can disagree in direction.

Data	Method	Family	Det.	Recall	Prec.	FA/day	Expo. %	Dup.	Utility
MetroPT	Single-sensor threshold	baseline	yes	1	0.000441	90.9	1.16	1	-452
MetroPT	Classical POT/GPD	baseline	yes	1	0.000396	101	1.60	3	-503
MetroPT	Nonlinear autoencoder	baseline	yes	1	0.00019	211	46.41	7	-1.05e+03
MetroPT	Conformal anomaly	baseline	yes	1	0.000396	101	2.80	3	-504
MetroPT	Empirical horizon risk	baseline	no	0	0	0	0.00	0	-2
MetroPT	Engineering threshold	baseline	yes	1	0.000645	62	75.10	4	-309
MetroPT	Ferro-Segers policy	baseline	yes	1	0.000393	102	0.12	3	-507
MetroPT	Fixed-run declustering	baseline	yes	1	0.000396	101	0.12	3	-504
MetroPT	Global empirical threshold	baseline	yes	1	0.000396	101	2.80	3	-504
MetroPT	Isolation forest	baseline	yes	1	0.000388	103	14.35	3	-514
MetroPT	K-gaps policy	baseline	yes	1	0.000394	102	0.12	3	-506
MetroPT	Linear autoencoder	baseline	yes	1	0.000212	189	7.24	5	-941
MetroPT	Regime-conditioned threshold	baseline	yes	1	0.000328	122	7.91	3	-608
MetroPT	Robust change point	baseline	yes	1	0.000387	103	14.64	3	-516
MetroPT	SPOT	baseline	no	0	0	1.16	0.04	0	-7.8
MetroPT	Negative-control region	negative control	no	0	0	12	0.53	0	-62
MetroPT	Recurrence score	registered	yes	1	0.00038	105	0.12	3	-525
MetroPT	Failure-prototype region	target region	no	0	0	0	0.00	0	-2
MetroPT	Rare-state region	target region	yes	1	0.000396	101	2.81	3	-504
MetroPT2	Single-sensor	baseline	yes	1	0.00282	21.2	7.38	3	-69.3

Table S3: Method provenance for the benchmark: score, threshold rule, event policy, alarm merge gap, parameter count, calibration status and tuning partition. Methods sharing a score are separated by their threshold rule or event policy; no two share all three.

Method	Score	Threshold rule	Event policy	Merge gap	Params	Calibration	Tuned on
Single-sensor threshold	best single sensor z	train quantile	merge gap	60	2	uncalibrated	train
Classical POT/GPD	robust max abs z	gpd return level	merge gap	60	3	uncalibrated	train
Nonlinear autoencoder	nonlinear projection residual	train quantile	merge gap	60	8	uncalibrated	train
Conformal anomaly	conformal tail probability	conformal level	merge gap	60	1	conformal (marginal, split)	train (calibration split)
Empirical horizon risk	Empirical horizon risk	train quantile	merge gap	60	2	empirical frequency	train
Engineering threshold	robust max abs z	fixed physical 3z	merge gap	60	1	uncalibrated	none (fixed a priori)
Ferro-Segers policy	robust max abs z	train quantile	ferro segers run length	0	2	uncalibrated	train
Fixed-run declustering	robust max abs z	train quantile	Fixed-run declustering	0	2	uncalibrated	train
Global empirical threshold	robust max abs z	train quantile	merge gap	60	1	uncalibrated	train
Isolation forest	Isolation forest	train quantile	merge gap	60	100	uncalibrated	train
K-gaps policy	robust max abs z	train quantile	k gaps run length	0	2	uncalibrated	train
Linear autoencoder	linear projection residual	train quantile	merge gap	60	8	uncalibrated	train
Regime-conditioned threshold	robust max abs z	regime conditioned quantile	merge gap	60	3	uncalibrated	train
Robust change point	absolute first difference	train quantile	merge gap	60	1	uncalibrated	train
SPOT	spot streaming excess	spot zero	merge gap	60	3	uncalibrated	train
Negative-control region	negative control distance	train quantile	merge gap	60	4	uncalibrated	train
Recurrence score	robust max abs z	train quantile	extremal index declustering	0	3	uncalibrated	train
Failure-prototype region	negative prototype distance	train quantile	merge gap	60	4	uncalibrated	train
Rare-state region	negative rare state distance	train quantile	merge gap	60	4	uncalibrated	train

Table S4: Fairness audit for the event-level benchmark. Shared protocol elements are checked for identity; event conversion is reported rather than required to be uniform, because several methods exist to vary it.

check	status	detail
configuration_hash	pass	one configuration_hash per dataset across all methods
target_events	pass	one target_events per dataset across all methods
leakage_control	pass	one leakage_control per dataset across all methods
no_test_partition_tuning	pass	no method declares test-partition tuning
method_result_distinctness	reported	methods with identical results (reported, not hidden): metropt: empirical_horizon_risk, failure_prototype_region; metropt: conformal_anomaly, fixed_run_declustering, global_empirical_threshold, rare_state_region; metropt2: conformal_anomaly, fixed_run_declustering, global_empirical_threshold; metropt2: ferro_seggers_event_policy, k_gaps_event_policy, rare_state_region
event_conversion_declared	reported	event policies in use: extremal_index_declustering, ferro_seggers_run_length, fixed_run_declustering, k_gaps_run_length, merge_gap; merge gaps in use: 0, 60
declared_specifications	pass	every reported method has a declared specification

4 Additional Real-Data Results

MetroPT, MetroPT2, SCANIA Component X, Hydraulic Systems, and SECOM differ in sampling, entity structure, labels, and exposure. The source registry records acquisition status and provenance for each dataset. MetroPT and MetroPT2 are compressor telemetry resources, SCANIA Component X is a fleet-level truck dataset with anonymized component histories, Hydraulic Systems is an experimental test-rig cycle dataset, and SECOM is a semiconductor process/yield dataset [5–9]. Because the datasets support different estimands and independent units, the industrial evidence is presented as a limitation and reproducibility target rather than a completed population-level comparison.

The event timeline artifact is retained because it can reveal aligned-axis mistakes, alarm bursts outside failure windows, and threshold behavior that aggregate tables may hide. It does not replace event matching, calibration, or independent-unit uncertainty.

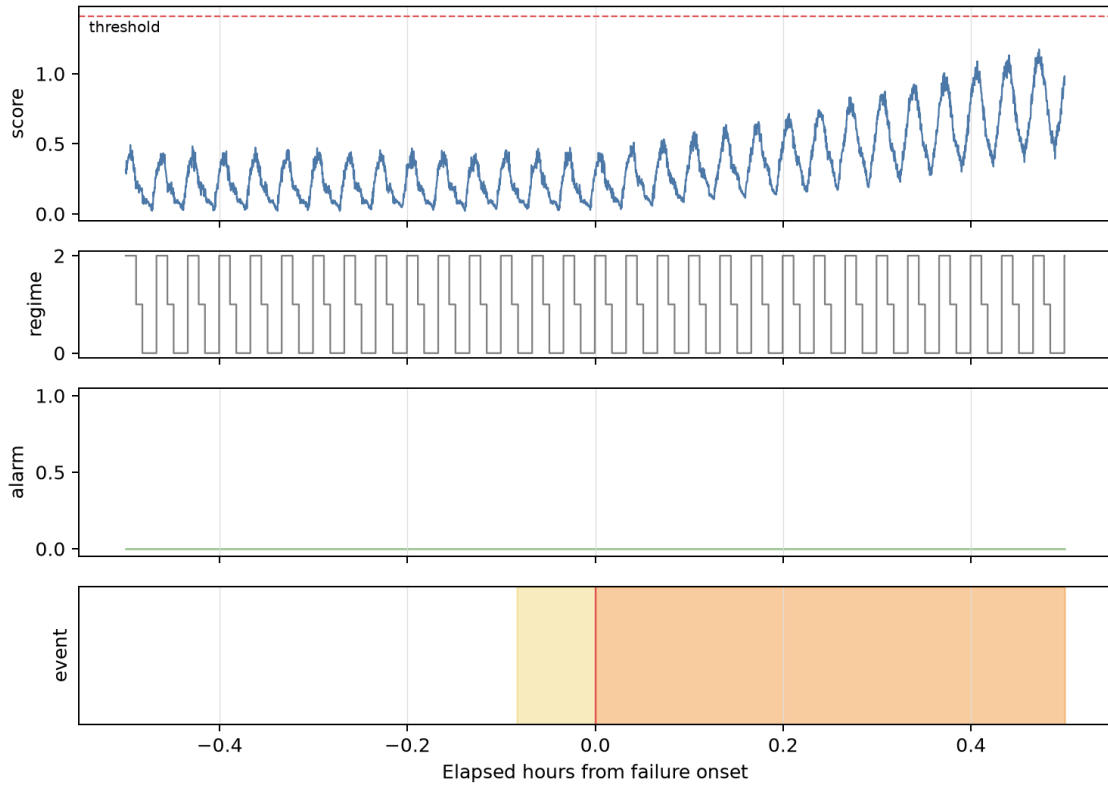


Figure S3: Supplementary alarm timeline generated from the cached processed data.

5 Robustness and Negative Results

The robustness policy is to preserve unfavorable outcomes. Threshold sensitivity, run-length sensitivity, regime-definition changes, dangerous-region definitions, missingness perturbations, and noise perturbations are all represented as registered families in the generated ablation table. Some cells are summaries of planned or non-estimable effects rather than completed industrial contrasts, so the ablation table is treated as a coverage record rather than a positive performance result.

Negative and null findings are part of the evidence. The manuscript does not turn diagnostic evidence into broad industrial performance claims, and it does not use empirical reliability

diagnostics as calibrated probability evidence. These constraints are enforced by the paper-check command and by generated audit artifacts.

5.1 Full Threshold and Alarm-Conversion Grids

Table S5: Failed-detection grid for threshold, recall, precision, and false-alarm burden.

threshold quantile	threshold	event recall	event precision	false alarm events per day	median warning lead time
0.9	1.119	0	0	168.5	
0.95	1.329	0	0	146.9	
0.97	1.388	0	0	133.9	
0.98	1.411	0	0	146.9	
0.99	1.439	0	0	138.2	

Table S6: Threshold and run-length stability diagnostics.

threshold quantile	run length	n exceedances	n clusters
0.9	0	2000	90
0.9	2	2000	41
0.9	5	2000	39
0.9	10	2000	38
0.9	20	2000	38
0.95	0	1000	105
0.95	2	1000	43
0.95	5	1000	34
0.95	10	1000	33
0.95	20	1000	33
0.97	0	600	141
0.97	2	600	44
0.97	5	600	31
0.97	10	600	31
0.97	20	600	31
0.98	0	400	157
0.98	2	400	48
0.98	5	400	34
0.98	10	400	31
0.98	20	400	31
0.99	0	200	128
0.99	2	200	68
0.99	5	200	32
0.99	10	200	31

5.2 Full Score-Stratification Bins

5.3 Ablation Configuration Registry

5.4 Baseline Runner Smoke Audit

Table S7: Full alarm-conversion grid for merge gap and post-onset matching tolerance.

merge	tol after	recall	prec	F1	false alarms	dup	FA/day	lead	warn time
0	0	0	0	0	157	0	678.2		400
0	30	0	0	0	157	0	678.2		400
0	60	0	0	0	157	0	678.2		400
0	120	0	0	0	157	0	678.2		400
0	300	0	0	0	157	0	678.2		400
5	0	0	0	0	34	0	146.9		586
5	30	0	0	0	34	0	146.9		586
5	60	0	0	0	34	0	146.9		586
5	120	0	0	0	34	0	146.9		586
5	300	0	0	0	34	0	146.9		586
20	0	0	0	0	31	0	133.9		604
20	30	0	0	0	31	0	133.9		604
20	60	0	0	0	31	0	133.9		604
20	120	0	0	0	31	0	133.9		604
20	300	0	0	0	31	0	133.9		604
60	0	0	0	0	31	0	133.9		604
60	30	0	0	0	31	0	133.9		604
60	60	0	0	0	31	0	133.9		604
60	120	0	0	0	31	0	133.9		604
60	300	0	0	0	31	0	133.9		604

Table S8: Split-aware score-stratification intervals using training-ranked scores.

split	lower	upper	count	pos	mean rank	obs prop	CI low	CI high
test	0.9	1	4000	401	1	0.1003	0.09132	0.1099
train	0	0.1	1199	0	0.05	0	0	0.003194
train	0.1	0.2	1200	0	0.15	0	0	0.003191
train	0.2	0.3	1201	0	0.25	0	0	0.003188
train	0.3	0.4	1199	0	0.35	0	0	0.003194
train	0.4	0.5	1200	0	0.45	0	0	0.003191
train	0.5	0.6	1201	0	0.55	0	0	0.003188
train	0.6	0.7	1200	0	0.65	0	0	0.003191
train	0.7	0.8	1199	0	0.75	0	0	0.003194
train	0.8	0.9	1200	0	0.85	0	0	0.003191
train	0.9	1	1201	0	0.95	0	0	0.003188
val	0	0.1	215	52	0.04765	0.2419	0.1895	0.3033
val	0.1	0.2	234	47	0.1494	0.2009	0.1545	0.2568
val	0.2	0.3	219	64	0.2529	0.2922	0.236	0.3556
val	0.3	0.4	280	126	0.352	0.45	0.3928	0.5086
val	0.4	0.5	292	86	0.4522	0.2945	0.2452	0.3492
val	0.5	0.6	283	119	0.5512	0.4205	0.3644	0.4787
val	0.6	0.7	366	197	0.651	0.5383	0.487	0.5887
val	0.7	0.8	323	164	0.7488	0.5077	0.4534	0.5618
val	0.8	0.9	317	134	0.8504	0.4227	0.3696	0.4777
val	0.9	1	1471	1311	0.9883	0.8912	0.8743	0.9061

Table S9: Ablation matrix summary.

ablation family	setting	summary value
threshold	0.9	0.1
threshold	0.95	0.05
threshold	0.97	0.03
threshold	0.98	0.02
threshold	0.99	0.01
run_length	0	157
run_length	2	48
run_length	5	34
run_length	10	31
run_length	20	31
lag	0	
lag	1	
lag	2	
lag	5	
lag	10	
regime	off	
regime	loaded	
regime	recovery	
dangerous_region	prototype	
dangerous_region	constraint	
dangerous_region	density	
noise_missingness	none	
noise_missingness	noise_0.05	
noise_missingness	missing_0.05	

Table S10: Synthetic smoke results for the shared baseline runner.

Baseline	Validation F1	Runtime (s)	Parameters
Engineering threshold	0	0.00142	12
Global empirical	0.911	0.00279	1
Regime empirical	0.892	0.0111	3
POT/GPD	0.911	1.77	3
Fixed-run declustering	0.904	0.00892	2
K-gaps declustering	0.904	0.00987	2
SPOT-style	0.919	14.8	4
Isolation Forest	0.92	2.63	120
Robust changepoint	0.566	0.0648	12
Linear autoencoder	0.955	0.0259	54
Conformal anomaly	0.319	0.0162	200
Empirical horizon risk	0.919	0.00306	200

6 Reproducibility

The build is organized around reproducible artifacts. The root command `make paper-assets` regenerates figures, tables, result macros, the results synthesis document, and per-file provenance records. The command `make paper` builds the main manuscript after regenerating assets. The command `make supplement` builds this supplement after the same asset step. The command `make paper-check` compiles both documents and audits generated assets, citations, compiled PDFs, and source markers. The command `make submission-package` assembles the reviewed manuscript package, reviewer report, revision matrix, availability statements, protocol, lock file, and artifact manifests without copying raw third-party datasets.

The generated macros provide stable result values used by the manuscript and supplement. Exact build identifiers and file-integrity records are stored in the repository manifests rather than printed in the narrative text.

Machine-readable interpretation and citation audits are written to `paper/generated/` by the paper checker. Manual edits to generated figures or tables are detected through the artifact-manifest audit.

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